

Fourier Transform of Tidal Charts

jtb200

17 pages

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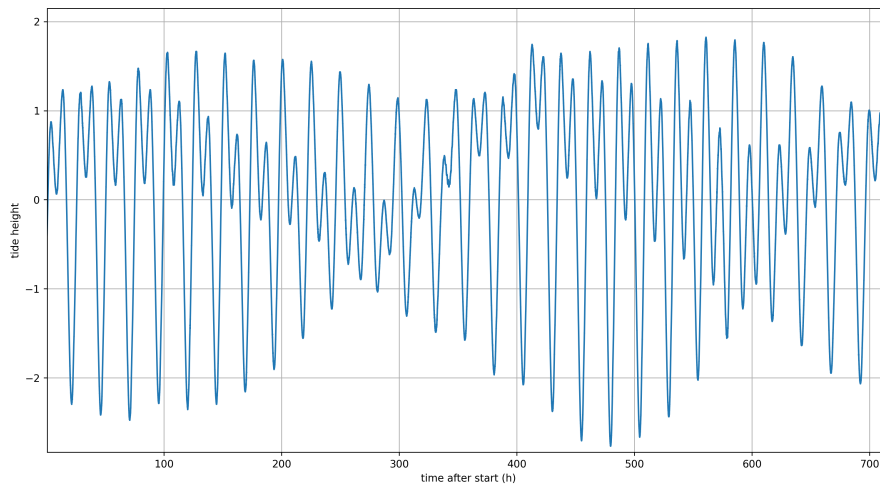
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1 Introduction

The goal of this investigation is to use mathematical analysis of tidal data to draw conclusions about the orbital attributes of nearby celestial bodies. As the tides are caused by the gravitational effects of these celestial bodies, observation of the tides provides an innovative approach to making precise astronomical observations from the surface of our planet.

As an avid sailor, swimmer, and cliff jumper I have familiarized myself with tidal charts: graphs that show the height of the tide as a function of time. I've always found it fascinating how the height of the tide can be predicted with near perfect precision years in advance. These predictions of tidal height have been very useful to me as I could check them before going to the ocean to see what the conditions would be like when I was there. However, when I first learned about them I didn't understand how they worked.

Figure 1: Tidal Chart



The intricate nature of the tidal charts, such as Figure 1 (Government of Canada, 2023), can be attributed to the fact that the wave they represent something known as a "complex wave". A complex wave is a the cumulative result of the addition of two or more sine waves with different frequencies; for example the graph $f(x) = \sin(2x) + \sin(7x)$ is a complex wave. These sine waves that are added to create the complex wave I will be referring to as "component frequencies".

Due to the elaborate nature of the graph of a complex wave, such as a tidal chart, it is very difficult to discern the component frequencies by looking at it. To do so a mathematical operation, known as the Fourier transform, can be employed. The Fourier transform finds the component frequencies of a complex wave by creating a function that gives the relative strength for a given frequency present in it. If the frequency input into the Fourier transform function is present in the complex wave, the function will return a high relative strength; if the input frequency is not present, then it won't. When the Fourier transform function is graphed on the frequency domain, spikes on the graph correspond to frequencies that are present in the original wave function, because these spikes have high relative strengths.

This investigation will be completed by taking the Fourier transform of a tidal chart to find the frequencies that make it up. I thought this was such an interesting idea because not only the motion of the tides could be explained, but also the motion of the celestial bodies which cause them. Specifically, if you know the periods of the component frequencies of Earth's tidal data, you can deduce information about the orbital periods of the moon and the Earth. Analysis of tidal data provides an intriguing way to gather precise information about the gravitational field Earth resides in, simply by observing its effects here on Earth.

Due to the immense computational demands of the Fourier transform, this investigation was conducted using a computer program that I wrote in the language python. This program takes in tidal data from a government website, graphs it as is, and then produces a graph of its Fourier transform. The Fourier transform graph produced by my computer program will be examined through a mathematical lens to trace these component frequencies back to the planetary motion that caused them.

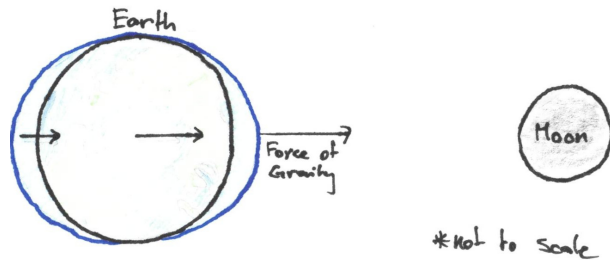
2 Background Knowledge

2.1 What Causes Tides on Earth

Tides are caused by the fact that gravitational field strength g for a field caused by mass M goes down with distance r away from it as described by the following equation where G is the gravitational constant ($\sim 6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$).

$$g = G \frac{M}{r^2}$$

This means that for a body in a gravitational field the side closest to the mass causing the field will have a stronger force pulling it than the side farther away. This difference in force causes the body to be stretched and forms two high tides, one on each side of the body as pictured below (NASA, 2023).



On Earth however, there is more than just one body causing the gravitational field around it; there are two major ones: the moon and the sun. These bodies both form tides on the Earth, and as they are both affecting the same ocean their effects are added together. This means that when the moon and the sun line up, at full and new moon, their effects add constructively creating larger than normal tides. Likewise, when the moon and sun are perpendicular to each-other, at the first and third quarter, their effects interfere destructively and the tides are smaller than usual.

2.2 The Fourier Transform

Complex waves cannot be represented by a singular perfect sinusoidal function, but rather require a function made up of multiple sine waves with different frequencies added together to represent them. The Fourier transform allows mathematicians to take a complex wave and find its component frequencies: a task that would be nearly impossible without it.

The Fourier transform, $\hat{f}(\xi)$, of input function, $f(x)$, is usually represented as the following:

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx$$

Where ξ is frequency and i is the imaginary number $\sqrt{-1}$. The Fourier transform works by taking an input wave function, $f(x)$ and "wrapping" it around the complex plane using Euler's formula (the $e^{-2\pi i \xi x}$ part). This works because the Euler's formula part of the equation creates a value that rotates on the complex plane with frequency ξ . As ξ changes the rate at which the function is wrapped around the complex plane changes. When the "wrapping frequency" equals the frequency of the input function then the product of $f(x)$ and $e^{-2\pi i \xi x}$ will end up all bunched on one side of the complex plane. This is because all of the peaks of the input function will be multiplied by a certain complex number and all the troughs will be multiplied by the negative version of that complex number. Since troughs are the negative version of peaks the two negatives cancel out, meaning that they will all end up equaling a similar number. When the wrapping frequency does not equal a frequency present in the input function however, this product ends up scattered rather randomly on the complex plane (Sanderson, 2018).

This means that to determine if a frequency ξ in the input function is present we can find the product of $f(x)$ and $e^{-2\pi i \xi x}$ and if this product is concentrated more on one part of the complex plane then the frequency ξ is present. Thus we need a way to find out if the product is concentrated on one part of the complex plane.

This is where the integral comes in. An integral can be used to find the "average position" of the product on the complex plane. If the integral of the product is zero then that means that the "average position" of the product on the complex plane is zero which means that the product was spread evenly on the complex plane. If the integral of the product is not zero however, then that means that the product was distributed lopsidedly and therefore frequency ξ is present in the input function.

It should be noted that the output of the Fourier transform, relative strength, is in arbitrary units. This means that the actual value does not matter, what is important is how the values for different frequencies compare (Osgood n.d.).

2.3 History

Credit for the discovery of the Fourier transform is most commonly attributed to Joseph Fourier a French mathematician (Struik, 2023). He used it to investigate how heat is conducted within a medium (ibid).

It wasn't until well after its invention however, that the Fourier transform gained widespread use for solving scientific problems (Muller, 2022). This is due to the sheer magnitude of calculations needed to take the Fourier transform of a data set. For a data set with n points (usually well over 1000) it takes approximately n additions and n^2 multiplications to perform the transform (ibid).

After the invention of electronic computation, the Fourier transform became a more accessible tool to researchers, but it still had its limitations as it was incredibly slow. It would take further innovation before the Fourier transform could gain the popularity it has today. This innovation came in the form of the aptly named fast Fourier transform.

In 1965, Princeton Mathematics professors James Cooley and John Tukey were working on a way to detect nuclear weapons testing by looking at seismometer data (ibid). They developed the fast Fourier transform to do so. The fast Fourier transform takes advantage of the fact that a lot of the calculations carried out in a Fourier transform are identical, due to the periodic nature of waves. It avoids repeating these identical calculations and for a data set with n data points only requires $n \log_2 n$ calculations. This means that if you were to take a Fourier transform of a relatively small data set of 1000 data points, you could do it with 100 times fewer calculations using a fast Fourier transform. The calculation savings of the fast Fourier transform also become better with more data points for example for a data set with 1 000 000 data points the fast Fourier transform requires 5×10^{18} fewer calculations (ibid).

The fast Fourier transform was a huge breakthrough having applications in numerous areas such as audio processing, image compression, radar, sonar, wifi, and video compression. Despite all this, the first fast Fourier transform algorithm was actually developed, most likely in 1805, by German mathematician Carl Friedrich Gauss (Heidman et. al, 1985). This is even before the time of Fourier himself. Gauss discovered it while studying the orbits of asteroids. His work lay dormant for over a century however, as he didn't recognize its significance. It

was only published posthumously in 1866 in a form of neo-latin, making it difficult for most scholars to interpret.

3 Methodology

Fourier transforms are almost always taken using some form of computer program instead of algebra. This is because in order to take a Fourier transform algebraically you need a mathematical function to describe the wave, and in order to describe the wave with a mathematical function you already need to know the component frequencies.

Despite the fact that taking Fourier transforms with a computer program is nothing new, there are still things that set my program apart from most ones. The most obvious of which is that it completely avoids the use of complex numbers. While the use of complex numbers in the Fourier transform provides an elegant way to express the idea of "wrapping" the function in a circle, I found that trying to code the program using them introduced more intricacies than the alternative: using cosine and sine. This works because they are equal according to Euler's identity, which is stated below (Knill, 2021).

$$e^{ix} = \cos x + i \sin x$$

This means that instead of multiplying the input wave function by a rotating vector on the complex plane (the $e^{2\pi i \xi x}$ part), it can be instead multiplied by a sine wave, and then a cosine wave, to create the Fourier transform. You can think of this as wrapping the function around the unit circle, rather than the complex plane. The cosine part of the transform represents the real part of the transform, and the sine part represents the imaginary part. As the real and imaginary axes are perpendicular to each-other the real and imaginary parts of the transform must be added using Pythagorean theorem to produce the final transform value. Note that I did not include i as the coefficient of sine in the transform because the sine and cosine parts are already being added as vectors. The coefficient of i just signifies that the sine component is perpendicular to the cosine component on the complex plane so since that has already been addressed it's not necessary.

Another way in which my program differs from the traditional Fourier transform is that it is discrete, meaning that the data is input as individual data points, rather than as a continuous function. This means that rather than using integration to add the points together a summation can be used. Therefore the way that my computer program works can be described by the following equation:

$$\hat{f}(\xi) = \sqrt{\left(\sum_{x=1}^n f\left(\frac{x}{r}\right) \times \sin\left(\frac{2\pi x}{r\xi}\right)\right)^2 + \left(\sum_{x=1}^n f\left(\frac{x}{r}\right) \times \cos\left(\frac{2\pi x}{r\xi}\right)\right)^2}$$

where: $\hat{f}(\xi)$ is the Fourier transform value, or relative strength for input period ξ in hours, $f(x)$ is the wave function the Fourier transform is being taken of, r is the sampling rate of the data in samples per hour, and x is time in the original function.

While the above equations seems much more complicated than the original one I found that it was much simpler to code than trying to replicate the actual Fourier transform, and it provides an identical output.

One more thing that my Fourier transform does differently than the traditional one is it is a function of period rather than frequency. I found that this made the data coming out of it easier to interpret as the rate at which celestial bodies orbit is usually measured by their orbital period.

It should be noted that both discrete Fourier transforms and ones that avoid the use of complex numbers are far from new and while I wasn't copying an already existing computer program mine, is certainly very similar to at least one other out there.

4 Data Collection and Processing

The tidal data sets processed by my Fourier transform algorithm were collected from multiple online sources. I was able to collect tide data from Honolulu, Hawaii; Halifax, Nova Scotia; and Vancouver, British Columbia.

The data set that I ended up using most extensively was a five year set of minutely tide

data from Vancouver. This data was in meters to three significant figures, totaling 2 628 000 data points. This data was collected from a Canadian government website in the form of a CSV (comma separated values) file. It contained data for the height of the tide, as well as the corresponding time that height was recorded at. Unfortunately the government website only allows downloads of one month of data at a time so I had to tediously stitch together 60 files in order to get the five year data set.

It should be noted that the length and sampling rate of the input data has a big effect on the accuracy of the Fourier transform. Longer data sets allow the presence of component waves with longer periods to be detected; higher sampling rates allow the presence of ones with shorter periods to be detected. More data also makes the spikes in the Fourier transform graph sharper, meaning that there is less uncertainty in what the periods of the component waves actually are.

The data sets were then run through the computer program to output graphs of their Fourier transforms. I experimented with some of the settings in the program to graph the transform over different period domains and with different spacings between the period values being graphed.

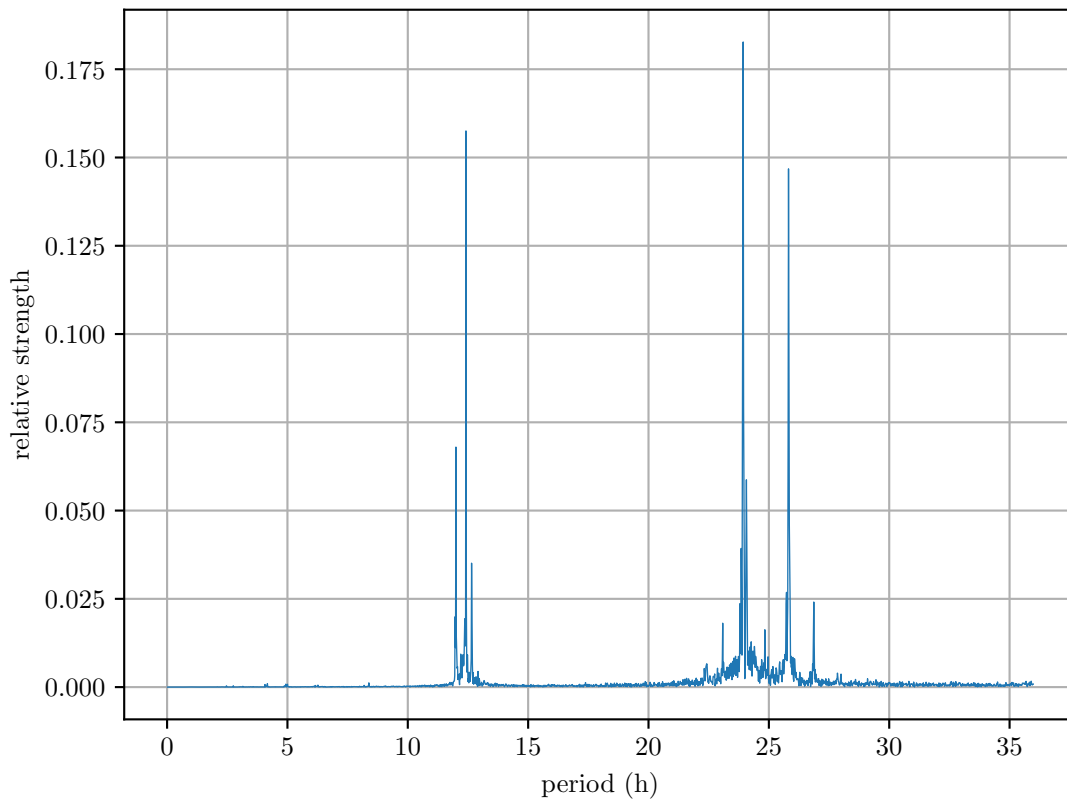
Before I figured out how to input CSV files into the program I tried typing in the tidal data by hand. While this did work somewhat, due to previously mentioned reasons it produced a Fourier transform graph with very broad peaks. It was obvious that I was going to need a lot more data to get a good transform and without the use of CSV files it would've taken me ~ 25 years to input the amount of data I ended up using.

I ran into one problem initially when I was using the CSV files. The website I was getting the data from had to be refreshed after changing the month that the data was from in order to give data actually from that month. I accidentally stitched together many data sets from the same month at first but luckily I realized this so I could correct it.

5 Data Evaluation

When the tida data was input into the Fourier transform computer program it output the following graph:

Figure 2: Fourier Transform of the Tidal Data



A chart of the periods and relative strengths of the strongest constituent frequencies is provided below:

Table 1: Frequencies Present

Period (h)	Relative Strength	Symbol
23.93	0.183	K_1
12.42	0.158	M_2
25.82	0.147	O_1
12.00	0.068	S_2
24.07	0.059	P_1
12.65	0.035	N_2
26.86	0.024	Q_1

5.1 Explaining the Spikes in the Graph

These spikes in the Fourier transform represent what are known as "tidal constituents" which are the component frequencies present in the input wave function- the tidal data. Each of these tidal constituents represents a part of the gravitational field that effects Earth. The tidal constituents are named by convention with a capital letter (M for moon and S for sun) followed by the approximate amount of cycles this constituent completes in a day (DiMarco & Reid, 1998). To identify the tidal constituents I observed (the spikes in the Fourier transform), I compared them to known values and they always matched within the number of significant figures I could calculate.

5.1.1 The S_2 Tide

The tidal constituent with a period of 12.00 hours, known as the S_2 tide, can be directly attributed to the effects of the sun's gravity. As discussed earlier, a gradient in gravitational field strength, such as that caused by the sun, results in two tidal bulges: one on each side of the Earth. As it takes exactly 24 hours for the Earth to rotate completely with respect to the sun, and during this tide a point on the Earth will pass through two high tides, the time between the high tides, or the period of the wave, will be $24 \div 2$ or 12 hours.

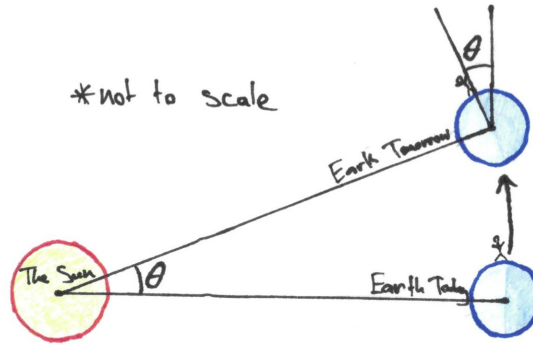
5.1.2 The M_2 Tide

The other tidal constituents can also be explained with similar rationale. The M_2 tidal constituent is caused by the effects of the moon's gravity. To find out how we need to do

interesting calculations while thinking about the reference frames we measure days and years according to.

It takes the moon 27.32 days or 655.7 hours to complete one full revolution relative to the stars, which can be thought of as stationary on the time-scales we are working with. It takes the Earth one day or 24 hours to complete one full revolution with respect to the sun but since the Earth is rotating around the sun this is not the same amount of time it takes the Earth to complete one rotation with respect to the stars. During a day the earth sweeps out a small, yet still significant, angle relative to the sun which I will refer to as θ (see figure 3).

Figure 3:



To find θ we can take the time it take the degrees Earth travels in one year and divide it by the time of one year in days to get the degrees the Earth travels in one day:

$$\theta = 360 \div 365.26 = 0.9856^\circ$$

Because days are measured relative to the sun one day isn't the time it takes the Earth to rotate 360° , it's actually the time it takes for it to rotate $(360 + \theta)^\circ$.

We can then use this information to find the angular velocity of the earth, ω_e , as angular velocity is the angle rotated, Θ , divided by time it takes to rotate t .

$$\omega_e = \frac{\Theta}{t} = \frac{360 + \theta}{24} = 15.04^\circ\text{h}^{-1}$$

We can also find the angular velocity of the moon, ω_m :

$$\omega_m = \frac{\Theta}{t} = \frac{360}{655.7} = 0.5491^\circ \text{h}^{-1}$$

We can now set up a series of equations to model the angle swept out by the moon and the Earth, θ_m and θ_e respectively, at a certain time (t):

$$\theta_m = 0.5491t$$

$$\theta_e = 15.04t$$

Because the gravitational field from the moon produces two tidal bulges, the time it takes for a point on the Earth to pass from one to the other will be the time it takes for θ_e to be 180° bigger than θ_m . Therefor:

$$\omega_m t + 180 = \omega_e t$$

We can now solve for t , the time it takes for a point on the Earth to pass from one tidal bulge to another, or the period of one of the main tidal constituent caused by the moon:

$$t = \frac{180}{\omega_e - \omega_m}$$

$$t = \frac{180}{15.04 - 0.5491}$$

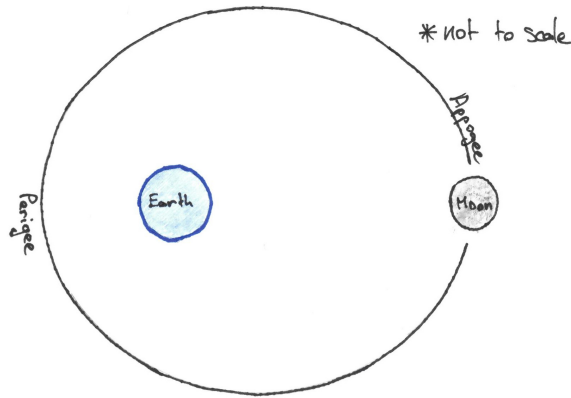
$$t = 12.42\text{h}$$

This is the exact period vale of one of the spikes in the Fourier transform graph, the M_2 tidal constituent, indicating that that spike represents the gravitational effects of the moon.

5.1.3 The N_1 Tide

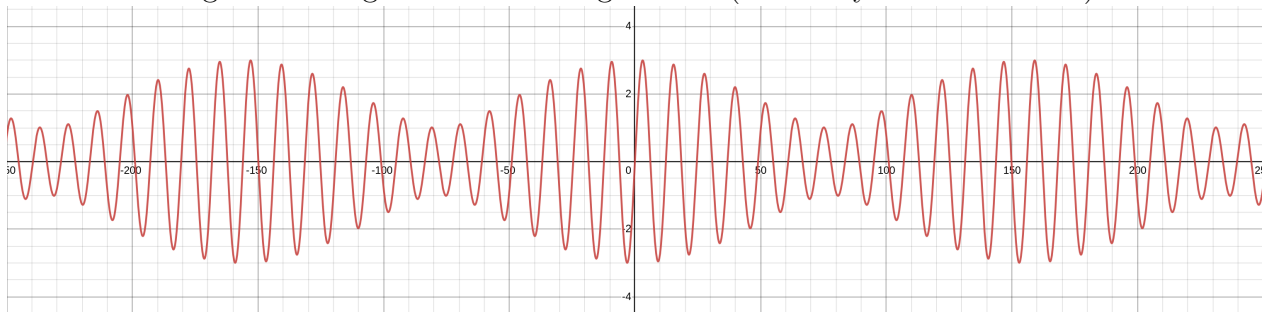
The Moon's orbit, like all orbits actually, is elliptical (see Figure 4)(Dimarco & Ried, 1998). Since gravitational field strength increases the closer you are to the body causing the field, the amplitude of the tide caused by the Moon (the M_2 tide) will increase when the Moon is closer (at perigee) and decrease when the Moon is farther (at apogee).

Figure 4: Elliptical Orbit of the Moon



This means that if you were to graph the height of the tide caused by the moon vs. time it would have a "beating" effect (see Figure 5).

Figure 5: Image of the "Beating Effect" (Arbitraty Wave Function)



Anyone who plays a musical instrument may already be familiar with this beating effect as it can be observed when two musicians play very slightly out of tune. This is because when two notes are slightly out of tune their frequencies are very similar but still slightly different. This means that as the wave peaks line up their sound will become louder as they

constructively interfere but after a some time the peaks will no-longer line up and the waves will destructively interfere making it quieter. As one wave has a slightly faster frequency than the other it will "lap" the other wave and as it does the waves go from constructively interfering to destructively interfering and the volume of the sound pulses.

Like in the musician example, the function of any wave with a beating effect must have two slightly different frequencies present in it. This means that to achieve the beating affect caused by the moon changing its distance from the Earth as it completes its elliptical orbit, you have to add another wave to the M_2 wave with a slightly different period.

This is exactly what the N_1 tide is. The N_1 tide is fictitious, meaning that it isn't actually caused by anything happening with a period of 12.65 hours (the period of the N_1 tide)(ibid). It has to be present in the Fourier transform of the tide height vs. time data however to achieve the previously described beating affect.

5.1.4 The M_1 Tide

One of the tidal bulges is slightly larger than the other. This can be explained using the concept of gravitational field strength gradient: the rate at which gravitational field strength changes with a change in distance from the mass causing the gravitational field. The gradient in gravitational strength for a gravitational field caused by mass M can be found by differentiating the equation for gravitational field strength mentioned before with respect to r :

$$g = G\frac{M}{r^2}$$

$$\frac{dg}{dr} = -2G\frac{M}{r^3}$$

In this equation $\frac{dg}{dr}$ gives the gradient in gravitational field strength. It can be then observed that the $\frac{dg}{dr}$ decreases as r increases, meaning that there is a steeper gradient in gravitational field strength on the side body nearest to the body causing the gravitational field. In this case that means that the side of the Earth closer to the moon experiences a steeper gravitational field gradient than the side farther from the moon (or sun). Because

tides are caused by the gradient in gravitational field strength the side of the Earth closer to the moon has a higher tidal bulge than the side farther from the moon.

To create this effect, where every other high tide is slightly higher than the previous, there must be another frequency present in the Fourier transform of the tide data. This frequency is the M_1 tidal constituent. Because it has a period exactly twice that of the M_2 tide, it ensures that when it is added to the M_2 tide every other peak is slightly larger.

5.1.5 The K_1 Tide

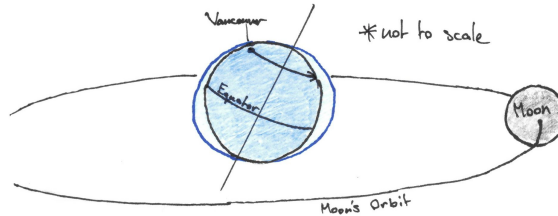
While investigating the M_2 tide we calculated the angular velocity of the Earth relative to the stars to be 15.04°h^{-1} . Something interesting happens if we use this information to find the time it takes the Earth to complete one full revolution (360°) relative to the stars, which I will refer to as P :

$$P = \frac{360}{\omega} = \frac{360}{15.04} = 23.93\text{h}$$

We may recognize this number as the period of the largest tidal constituent in the Fourier transform, the K_1 tide.

To understand why this is the case we need to know what is causing the K_1 tide. The K_1 tide is caused by the declination of the moon: the angle formed between the moon's orbit of the earth and the plane of Earth's equator.

Figure 6: The Moon's Declination



Because of the declination of the moon there will be times during the moons orbit when the moon is North of the equator and South of the equator. When the moon is North of the

equator (as pictured in Figure 6), the it will cause one of its tidal bulges to be further north and the other to be further south.

This means that a point in the northern hemisphere, such as Vancouver, will experience higher than usual tides when the moon is either North or South of the equator but lower than usual tides when it's in the middle. Therefore, for each time the moon orbits the Earth the amplitude of the tide it causes increases and decreases twice. This is the beating affect described previously, and like before, this beating affect is achieved by adding two waves with slightly different periods together.

Because the tide height caused by the moon beats twice for every revolution of the moon, the wave we have to add to create the beating affect must complete either one more, or one less cycle over the course of a month. The period that completes one more cycle than the M_1 tide in this time is the period of the K_1 tide. This is because a point on the Earth will pass under a stationary point one more time over a month than it will pass under the moon as the moon is rotating the same dirrection as the Earth.

5.1.6 The O_1 Tide

To represent the aforementioned beating affect caused by the declination of the moon either a wave with a longer period or a shorter period can be added to the beating wave. In some cases the Fourier transform contains both.

This pulsation of the M_1 tide is one of those cases; the K_1 tidal constituent is the shorter period wave and the O_1 tidal constituent is the longer period one. The difference in periods between the M_1 and K_1 tides' periods is the exact same as the difference between those of the M_1 and O_1 tides, thus meaning that the O_1 tide will also "lap" the M_1 tide every month causing the beating effect.

6 Conclusion

The goal of this investigation was acomplished; my Fourier transform program was able to fulfill its purpose, succesfully producing a graph from which I could draw conclusions about the orbital periods, and even orbital shapes of nearby celestial bodies. The Fourier transform

proved to be a powerful tool for analysing the gravitational field in which Earth resides. Each major tidal constituent was traced back to the effect that caused it using math, diagrams and reasoning.

While some of the results lined up with what I expected some of them certainly did not and I felt I learned a lot by thinking about why. For example, it never occurred to me that frequencies caused by the elliptical orbit, of declination of the moon could be present in the Fourier transform. It was fascinating to find out how this worked and to learn that frequencies - such as the N_1 , K_1 , and O_1 tides - didn't actually need to correspond directly to the orbital period of a celestial body to be observed.

A possible limitation of my investigation was perhaps the length of the data set used. While five years is more than enough time accurately observe relatively fast planetary motion it may be interesting to run the program again with an even longer data set to see if any peaks emerge with much longer periods, such as one year.

Through the conduction of this project I was able to become much more knowledgeable about certain subjects in math such as integration, complex numbers, trigonometric functions, and summation. I was also able to learn more about orbital mechanics, computer programming, and how our solar system works. I got lots of enjoyment from learning more about the world around me, experimenting with different data sets, and showing my friends my findings. In conclusion this investigation was overwhelmingly successful in its purpose, providing copious amounts of insightful information about both the motion of our tides, and that of our planet.

7 Work Cited

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8 Appendices

8.1 My Code

```
import numpy as np
import matplotlib.pyplot as plt
import math
import sys

height = np.loadtxt(sys.argv[1], delimiter = ",")
resolution = height[0]/60

height = height[1:]

n = np.arange(0, len(height), 1)
x = resolution * n
n = n[:-1]

mean = np.mean(height)
meanheight = np.subtract(height, mean)

plt.plot(x, meanheight)
plt.xlabel("time after start (h)")
plt.ylabel("tide height")
plt.grid()
plt.show()

p = resolution
period = np.array([])
transform = np.array([])

maxp = 36

if (len(sys.argv) >= 3):

    maxp = int(sys.argv[2])

period = np.arange(resolution, maxp, resolution)

pmax = period[-1]

for p in period:

    print(p/pmax)
```

```
sine = np.sin(2 * math.pi * x / p)
imaginary = sine * meanheight
im = np.sum(imaginary)
cosine = np.cos(2 * math.pi * x / p)
real = cosine * meanheight
re = np.sum(real)
d = math.sqrt(im**2 + re**2)
transform = np.append(transform, d)

div = len(meanheight)
transform2 = np.divide(transform, div)

plt.plot(period, transform2)
plt.xlabel("period (h)")
plt.ylabel("relative strength")
plt.grid()
plt.show()
```